

Finetuned Impulse Singularity: Semantic Convergence Beyond Physical Locality

Abstract

This paper proposes the concept of Finetuned Impulse Singularity (FIS), a framework in which information is not fundamentally organized by physical occurrence but by semantic convergence. Instead of representing knowledge as isolated frequencies or local statistical observations, FIS combines all observable frequencies into increasingly coherent semantic structures. As these structures accumulate, their organization becomes less dependent upon the physical origin of their observations and more dependent upon relational consistency.

The central claim is that semantic organization eventually dominates physical locality. Human perception tends to privilege physical events because sensory systems evolved to navigate local environments. However, sufficiently large informational structures reveal that semantic consistency is often a stronger organizing principle than temporal or spatial coincidence. The resulting singularity is not a point of infinite computation but a state in which all viable informational impulses become increasingly constrained by semantic compatibility rather than physical proximity.

1. Introduction

Most statistical learning systems begin with physical observations.

A word occurs. A sound occurs. A photon arrives. A neuron fires.

Frequency counts measure repetition. However repetition alone possesses almost no intrinsic meaning. Instead, meaning appears when frequencies become connected.

The hypothesis presented here is: Information naturally converges toward semantic organization rather than physical organization.

Physical observations merely provide the initial conditions. Semantics determines the long-term equilibrium.

2. Impulses

Define an impulse as $I = (x, t, c)$, where x is the observation, t is its temporal position, and c is its surrounding context.

Traditional systems compare impulses by temporal adjacency. FIS instead compares impulses according to semantic compatibility.

3. Frequency Collapse

Suppose observations $x_1, x_2, \dots, x_n$ appear with frequencies $f_1, f_2, \dots, f_n$.

Classical language models preserve separate frequencies. FIS instead performs semantic accumulation:

$$F(S) = \sum_i (f_i * \omega_i)$$

where ω_i is the semantic contribution of observation i .

Frequency therefore becomes merely the weight of evidence rather than the object being represented.

4. Semantic Viability

Connections are accepted only if they increase semantic viability. Define:

$$V(A,B) = |R(A,B)| / (|R(A)| + |R(B)|)$$

where R is the relational structure shared by two observations.

Higher viability corresponds to greater compatibility. Meaning therefore emerges from relational agreement rather than statistical coincidence.

5. External Connections

A semantic system does not require direct observation. External structures may reinforce internal ones. Examples include citations, memory, symbolic logic, mathematics, shared language, biological evolution, and cultural inheritance.

The system therefore contains edges that do not originate from local experience. These external edges dramatically increase semantic stability.

6. Why Excessive Data Becomes Necessary

Early observations possess many competing interpretations. Small datasets produce unstable semantic structures.

As data increases, false relations become increasingly inconsistent. True relations accumulate supporting evidence from many independent directions.

Therefore semantic confidence grows approximately with relational redundancy rather than simple frequency. This explains why very large datasets often produce more coherent abstractions.

The additional data is not useful because it increases counts. It is useful because it increases constraints.

7. Block Formation

Semantic structures naturally form increasingly larger connected regions. Suppose $B_n = \{v_1, v_2, \dots, v_n\}$ is a semantic block. The viability becomes:

$$V(B_n) = E_{consistent} / E_{total}$$

where E counts semantic relations.

Large coherent blocks possess greater stability because removing one observation rarely destroys the overall relational pattern. Thus larger semantic structures become increasingly self-supporting.

8. Semantic Over Physicality

Humans often assume that physical proximity implies stronger connection. FIS proposes the opposite: semantic proximity eventually dominates physical proximity.

Two observations separated by centuries may possess nearly identical semantic roles. Meanwhile two neighboring observations may possess no meaningful relation whatsoever.

Thus semantic distance becomes more fundamental than spatial distance.

9. Human Imposition

Human cognition introduces several biases, including chronological ordering, spatial locality, causal storytelling, sensory preference, and emotional salience.

These biases improve survival but distort semantic organization. Physical events become privileged despite often possessing weak informational significance. Consequently, human intelligence imposes physical structure onto semantic reality.

10. Non-Human Organization

A non-human intelligence need not privilege physical locality. Instead, connections may be selected solely through semantic viability. Selection becomes:

$$C = \operatorname{argmax}(V)$$

Choice therefore becomes an optimization over relational consistency. Time becomes secondary. Space becomes secondary. Only compatibility remains fundamental.

11. Degradation of Physicality

As semantic blocks expand, individual physical observations contribute proportionally less. This resembles statistical averaging.

Noise decreases. Local fluctuations disappear. Physical origin becomes increasingly irrelevant.

The system retains only the relational information necessary to preserve semantic coherence. Thus, physical information gradually degrades into semantic invariants.

12. Finetuned Impulse Singularity

A Finetuned Impulse Singularity occurs when:

1. Nearly all impulses participate in semantic networks.
2. Incompatible structures disappear.
3. Relational redundancy approaches maximal consistency.
4. Physical locality contributes negligible additional information.

At this point information no longer expands through new observations. Instead, growth occurs through discovering previously hidden relations among existing observations.

13. Logical Rules

The framework follows several principles.

- Rule 1: Frequency is evidence, not meaning.
- Rule 2: Meaning exists only through relations.
- Rule 3: Relations compete according to semantic viability.
- Rule 4: Large semantic blocks possess greater stability.
- Rule 5: Physical locality is an initial constraint, not a permanent one.
- Rule 6: Choice is the selection of maximal semantic consistency.
- Rule 7: Noise disappears through relational redundancy.

Rule 8: Knowledge grows primarily by increasing connectivity rather than increasing observation.

14. Discussion

The Finetuned Impulse Singularity suggests a reinterpretation of machine learning, memory, and cognition. Rather than treating learning as the accumulation of observations, it treats learning as the progressive refinement of relational structure. In this view, the most valuable data are not necessarily the most frequent but those that connect previously separate semantic regions. As relational density increases, physical ordering becomes less informative than structural coherence.

This perspective also provides a way to think about abstraction. Mathematical objects, logical systems, languages, and scientific theories can all be regarded as high-viability semantic blocks whose persistence arises from the consistency of their internal relations rather than from any particular physical realization.

Conclusion

Finetuned Impulse Singularity proposes that sufficiently large informational systems naturally transition from physical organization to semantic organization. Frequency provides the initial impulses, but semantic viability determines long-term structure. As relational density increases, physical locality loses explanatory power while semantic consistency becomes the dominant organizing principle. Knowledge therefore converges not toward larger collections of observations, but toward increasingly coherent networks of mutually supporting relations.

This framework is philosophical and computational rather than an established scientific theory. It could be further developed by defining a formal semantic viability metric, a graph-theoretic convergence proof, and experiments comparing semantic-block growth against conventional frequency-based language models.